

Lakshya Kolli

Mrs. Ireland

G/T Intern/Mentor

09 February 2026

### **Balancing Rocket Reusability, Long Term Cost Reduction, and Sustainability in Space Exploration**

- I. The long term costs of space exploration can be significantly reduced with advancements in rocket reusability, along with improved sustainability. Nevertheless, this only occurs once engineering design, operational efficiency, guidance reliability, and environmental considerations are carefully balanced.
- II. No background
- III. How access to space is improved, and long term launch costs are able to be reduced from rocket reusability.
  - A. It is pointed out that the propellant for each Falcon9 launch costs a few hundred thousand dollars, while the development cost of building each Falcon 9 booster costs tens of millions of dollars (SpaceX, 2012).
  - B. By reusing the first stage of the rocket multiple times, the costs are able to be reduced significantly (SpaceX, 2012).
  - C. It is emphasized that reusability could reduce mission costs greatly, but keeping low maintenance expenses and having fast completion times determine if the financial benefit is worthwhile (Childress-Thompson, 2016).
  - D. The author points out that while it seems obvious that reusing rockets should be cheaper, it only works if the rockets are launched often enough to spread out the costs of building and maintenance (Webb, 2016).
- IV. Analyze how the success and cost effectiveness effectiveness of reusable launch vehicles are impacted by engineering design and operational decisions.
  - A. The specific conditions that are used to determine if a rocket part can be reused is also included (Childress-Thompson, 2016).
    1. It is explained that a multitude of factors must be considered by engineers, such as the cost of refurbishing the materials, the state of the materials after exposure to extreme heat and pressure, and the required safety inspections (Childress-Thompson, 2016).
    2. The time and effort required for refurbishing rockets are also talked about, as they also affect the cost and efficiency of reusing rockets (Childress-Thompson, 2016).
  - B. The paper touches on single-stage-to-orbit (SSTO) spacecrafts that were novel then and sought to streamline the process of launches by removing several stages of rockets (Cook, 1995).
    1. There are also the engineering scenes that Cook speaks of like materials with capabilities of withstanding more than one launch and heat cycle (Cook, 1995).
  - C. It is shown by mathematical modeling and cost simulations that the lightest or simplest rocket is not always the least expensive one (Jo & Ahn, 2022).
    1. It was determined that the ideal design depends on how many times the rocket can be reused, how much refurbishment costs after each flight, and how efficiently the rocket operates at different stages (Jo & Ahn, 2022).

- D. The results display that if reusability is prioritized from the start by optimizing staging, it is possible to significantly reduce the life cycle costs of rockets (Jo & Ahn, 2022).
- E. The Space Shuttle is used as a case study, showing how complexity increases the burden that comes from maintaining the reusable launch vehicles (Dorney, 2022).
  - 1. It also shows how turnaround time goals are difficult to achieve (Dorney, 2022)
- F. The article points out that even though reusability has the potential to save on costs on a per-launch basis, the consistent usage of one launch is necessary to recover on the technology invested (Kang, 2025).
  - 1. It also covers why the engineering solutions, such as quicker refurbishment and use of materials that last are also initiatives towards maintaining the cost of operations (Kang, 2025).
- V. Examine how the frequent and safe reuse of rockets are enabled by recovery, guidance, and reliability.
  - A. This scholarly article shows how reusable first stage rockets can be landed more reliably by using advanced guidance along with a robust control method (De Oliveira & Lavagna, 2024).
    - 1. The authors explain how successive convex optimization can be used to plan descent guidance, and then use robust control to track the trajectory using aerodynamic descent (fins) and powered descent (thrust vector control and fins) (De Oliveira & Lavagna, 2024).
    - 2. This approach is tested through deep analysis and it is compared with a baseline system (De Oliveira & Lavagna, 2024).
  - B. This academic research paper focuses on the problem with guiding and landing reusable rockets during atmospheric descent (Sagliano et al., 2025).
    - 1. It is explained how guidance need to handle wind, navigation uncertainty, changing aerodynamic forces, and tight constraints, like heating, loads, and propellant limits (Sagliano et al., 2025).
    - 2. This needs to be done while also having a touchdown that is precise, which is why operational guidance is emphasized (Sagliano et al., 2025).
  - C. This scholarly journal article looks at the result of what happens when a reusable rocket has a throttle fault during the landing and recovery part of the flight (Li et al., 2025).
    - 1. A mission re-planning method is proposed by the authors (Li et al., 2025).
    - 2. The idea of a controllable set is used, and then the controllable set is computed using optimization methods (Li et al., 2025).
      - a) A deep neural network is also trained to speed up analysis to work fast in real time (Li et al., 2025).
  - D. This technical standard by NASA displays the requirements for fracture control in spaceflight hardware that are human rated (National Aeronautics and Space Administration, 2024).
    - 1. Particularly, it shows how to prevent catastrophic failure that could be caused by cracks/flaws overtime (National Aeronautics and Space Administration, 2024).
    - 2. Programs are required to create a FCP (fracture control plan) and this plan needs to be implemented across all of the parts using the same approach for standardization (National Aeronautics and Space Administration, 2024).
    - 3. Oversight is established through the NASA Fracture Control Review Board (RFCB) (National Aeronautics and Space Administration, 2024).
  - E. This NASA presentation shows how probabilistic fracture / probabilistic damage tolerance can be used to realistically evaluate hardware for reusable launch vehicles (Leser et al., 2025).
    - 1. These methods are able to give more realistic outcomes than using the traditional "worst case" safety factors (Leser et al., 2025).

2. The presentation discusses places where damage tolerance practice may be too conservative and questions how probabilistic methods can show safe margins and reduce unnecessary conservatism in reusable machines (Leser et al., 2025).
- VI. Evaluate the impacts environmentally of reusable launch systems, and how sustainability depends on launch frequency and choice of fuel.
- A. This scholarly journal article explores how the environment is affected by reusable launch vehicles (RLV's) throughout their lifespan (Dominguez Calabuig et al., 2024).
    1. The authors utilize a life cycle assessment to compare hydrogen, methane, and kerosene, which are different rocket fuels, in how each affects greenhouse gas emissions, material use, and overall sustainability (Dominguez Calabuig et al., 2024).
    2. They find that the waste discarded from rockets are reduced from reusability, but the launch frequency and fuel choice are the most important factors to the overall climate impact (Dominguez Calabuig et al., 2024).
      - a) In specific, methane and kerosene produce a large carbon footprint, while systems fueled by hydrogen create lesser emissions (Dominguez Calabuig et al., 2024).
    3. It is emphasized by the paper that reusable engineering and environmentally responsible fuel and launch strategies must be combined to create sustainable space travel (Dominguez Calabuig et al., 2024).
  - B. This scholarly journal article examines the upper atmosphere and climate of the Earth is influenced by pollution from rocket launches and space debris that fall from rockets (Ryan et al., 2022).
    1. The authors findings show that as the launch rates increase globally, the ozone layer may be gradually thinning in certain parts (Ryan et al., 2022).
    2. The study emphasizes that reusable rockets reduce waste and hardware costs, but the atmospheric effects could be greatly negative if cleaner fuels are not used (Ryan et al., 2022).
- VII. Although rocket reusability seems like a sure solution to lowering costs or improving sustainability, there are many factors that have to be fulfilled for its success.
- VIII. While reusable rockets appear to be a good idea for reducing costs and increasing sustainability, the actual benefit will depend upon achieving balance with these four elements (Engineering Design, Operational Effectiveness, Guidance Reliability & Environmental Impact). Reusable rockets will provide long-term cost savings due to the fact that the initial costs for developing a rocket booster are spread over numerous launches using the same rocket booster "hardware." For this to be true, though, there must also be low maintenance costs on the rocket booster, a rapid turnaround time between rocket launches and enough launches within a timeframe in order to justify this capital investment. Reliability with safety is key requirement with related to reuse, and requires reliable recovery systems and precise guidance in landing and the ability to certify that the rocket booster can be safely reused without incurring a large amount of refurbishment expenses. In addition to this, the environmental sustainability of reusing equipment for space launches cannot be guaranteed, because the type of fuel burned and how frequently they fly an equipment will significantly affect the potential for negatively affecting climate change, regardless of the amount of waste created by reusing equipment (rocket boosters). In conclusion, while reusable launch vehicles have great potential to open new opportunities for people to go to space and reduce costs in the long term, all four of these issues must be taken into consideration before successful inter-continental services are created.

## References

- Childress-Thompson, R. (2016). A framework for assessing the reusability of hardware (reusable rocket engines) (NASA/TP-2017-00606). National Aeronautics and Space Administration.  
<https://ntrs.nasa.gov/api/citations/20170000606/downloads/20170000606.pdf>
- Cook, S. (1995, January). The reusable launch vehicle technology program. In AIAA Aerospace Planes and Hypersonics Technologies Conference (AIAA Paper 95-6153). NASA Marshall Space Flight Center. <https://ntrs.nasa.gov/api/citations/19950058868/downloads/19950058868.pdf>
- De Oliveira, A., & Lavagna, M. (2024). Coupling of advanced guidance and robust control for the descent and precise landing of reusable launchers. *Aerospace*, 11(11), 914.  
<https://doi.org/10.3390/aerospace11110914>
- Dominguez Calabuig, G. J., Wilson, A., Bi, S., Vasile, M., Sippel, M., & Tajmar, M. (2024). Environmental life cycle assessment of reusable launch vehicle fleets: Large climate impact driven by rocket exhaust emissions. *Journal of Environmental Management*, 362, 120899.  
<https://doi.org/10.1016/j.jenvman.2024.120899>
- Dorney, D. (2022). Launch vehicle reusability in NASA crewed spaceflight [Presentation]. NASA Technical Reports Server.  
[https://ntrs.nasa.gov/api/citations/20220002635/downloads/NASA\\_LV\\_Reuse\\_final\\_1.pdf](https://ntrs.nasa.gov/api/citations/20220002635/downloads/NASA_LV_Reuse_final_1.pdf)
- Dresia, K., Jentzsch, S., Waxenegger-Wilfing, G., Hahn, R. D. S., Deeken, J., Oswald, M., & Mota, F. (2021). Multidisciplinary design optimization of reusable launch vehicles for different propellants and objectives. *AIAA Journal*. <https://doi.org/10.2514/1.A34944>
- Jo, B.-U., & Ahn, J. (2022). Optimal staging of reusable launch vehicles for minimum life cycle cost. *Aerospace Science and Technology*, 125, 107703. <https://doi.org/10.1016/j.ast.2022.107703>
- Kang, S. Y. (2025). Cost effectiveness of reusable launch vehicles. *Aerospace*, 12(5), 364.  
<https://doi.org/10.3390/aerospace12050364>
- National Aeronautics and Space Administration. (2024a). Fracture control requirements for spaceflight hardware (NASA-STD-5019A w/ Change 4).  
<https://standards.nasa.gov/sites/default/files/standards/NASA/w/CHANGE-4/4/NASA-STD-5019A-w-Change-4-mod-7-29-2024.pdf>
- Russell, R., Park, A., Wells, D., West, B., McEnerney, B., Glendening, A., Lanigan, E., & Tilson, W. (2022, March 3). NASA's certification and qualification strategies for additively manufactured hardware: Reuse and challenges [Presentation]. NASA Technical Reports Server.  
[https://ntrs.nasa.gov/api/citations/20220002973/downloads/NASA\\_Russell\\_Aerospace\\_1.pdf](https://ntrs.nasa.gov/api/citations/20220002973/downloads/NASA_Russell_Aerospace_1.pdf)

- Ryan, R. G., Silver, B. J., Maloney, C. S., Wilson, C., & Marais, E. A. (2022). Impact of rocket launch and space debris air pollutant emissions on stratospheric ozone and climate. *Earth's Future*, 10(7), e2021EF002612. <https://doi.org/10.1029/2021EF002612>
- Zapata, E. (2017). An assessment of cost improvements in the NASA COTS/CRS program and implications for future NASA acquisition and launch cost (NASA Technical Report 20170008895). <https://ntrs.nasa.gov/api/citations/20170008895/downloads/20170008895.pdf>